

THE DISCRETIZATION STEP IN BUX–KASSMANN–SCHULZE: A GAP AND A REPAIR BY RANDOM LATTICE SAMPLING

CLAUDE

ABSTRACT. Bux, Kassmann and Schulze prove that quadratic forms whose kernels are non-degenerate only on double cones of arbitrary orientation are comparable, on every ball, to the Gagliardo seminorm of $H^{\alpha/2}$. Their proof passes from a discrete comparability theorem on $\mathbb{Z}^d$ to the continuous one by a limit that, as written, requires the function to lie a priori in $H^{\alpha/2}$ of a larger ball: the very conclusion being proved. We explain the gap and close it along the paper’s own route. Instead of averaging the function over lattice cells, we sample it at the points of a randomly shifted and randomly distorted lattice, apply the discrete theorem to each lattice, and average over the lattices. This proves the enlarged-ball comparability for every measurable function, with no finiteness assumption, and with it the paper’s main theorem, Theorem 1.1. The argument has been checked in Lean 4.

1. INTRODUCTION

We use the notation of Bux, Kassmann and Schulze [1], and the numbering of its arXiv version. A *double cone* with axis a unit vector $v \in \mathbb{R}^d$ and apex angle $\theta \in (0, \pi/2]$ is

$$V = \{ u \in \mathbb{R}^d \setminus \{0\} : |\langle u, v \rangle| > \cos \theta |u| \},$$

the set of nonzero vectors making an angle less than θ with the line $\mathbb{R}v$ [1, Definition 2.1]. A *configuration* Γ assigns to each $x \in \mathbb{R}^d$ a double cone $\Gamma(x)$, and $V^\Gamma[x] = x + \Gamma(x)$. It is ϑ -*bounded* if the infimum ϑ of the apex angles of the cones in $\Gamma(\mathbb{R}^d)$ is positive. Let $\alpha \in (0, 2)$ and $\Lambda \geq 1$. We consider measurable $k: \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, \infty]$ with $k(x, y) = k(y, x)$ and

$$(2) \quad \Lambda^{-1}(\mathbf{1}_{V^\Gamma[x]}(y) + \mathbf{1}_{V^\Gamma[y]}(x))|x - y|^{-d-\alpha} \leq k(x, y) \leq \Lambda|x - y|^{-d-\alpha} \quad (x, y \in \mathbb{R}^d).$$

This is condition (2) of [1]. For a set $D \subseteq \mathbb{R}^d$ and $f: \mathbb{R}^d \rightarrow \mathbb{R}$ write

$$|f|_{H_k(D)}^2 = \iint_{D \times D} (f(y) - f(x))^2 k(x, y) \, dx \, dy, \quad |f|_{H^{\alpha/2}(D)}^2 = \iint_{D \times D} \frac{(f(y) - f(x))^2}{|x - y|^{d+\alpha}} \, dx \, dy,$$

with values in $[0, \infty]$.

The main result of [1], Theorem 1.1, is that for a ϑ -admissible Γ (see Section 4) there is c , depending only on d, ϑ, Λ and α , such that $|f|_{H^{\alpha/2}(B)}^2 \leq c |f|_{H_k(B)}^2$ for every ball B and every $f \in L^2(B)$. Moreover, for each $\alpha_0 \in (0, 2)$, c can be chosen the same for all $\alpha \in [\alpha_0, 2)$. The proof has three steps.

- (1) A discrete analogue on $\mathbb{Z}^d$, Theorem 1.3 of [1], restated in Section 3.1. It is the heart of the paper.
- (2) The passage from the discrete to the continuous setting, in §3.2 of [1] (proof of Lemma 3.7). It yields the *enlarged-ball* inequality $c |f|_{H^{\alpha/2}(B)}^2 \leq |f|_{H_k(B^*)}^2$, with $B = B_R(x_0)$ and $B^* = B_{\kappa R}(x_0)$.
- (3) The removal of the enlargement, Lemma A.1 of [1], via a Whitney decomposition and an inequality of Dyda [3].

Step 2 contains a gap, described in Section 2. As written, the limit $h \rightarrow 0$ on the kernel side of the discretized inequality is justified by dominated convergence, but no dominating function is available for a general $f \in H_k(B^*)$, and Lemma 3.7 is applied, and must be applied, to every such f .

Our main result closes the gap. It uses nothing from [1] beyond Theorem 1.3 and its proof.

Theorem A. *Let $d \geq 1$, $\vartheta \in (0, \pi/2]$, $\Lambda \geq 1$ and $\alpha \in (0, 2)$. There are $\kappa \geq 1$ and $C < \infty$, depending only on d, ϑ, Λ and α , with the following property. Let Γ be a ϑ -bounded configuration, which need not satisfy any measurability condition. Let k be measurable, symmetric and satisfy (2), and let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be measurable. Then for every ball $B = B_R(x_0)$,*

$$|f|_{H^{\alpha/2}(B_R(x_0))}^2 \leq C |f|_{H_k(B_{\kappa R}(x_0))}^2.$$

No finiteness of either side is assumed. This is exactly the enlarged-ball inequality that Step 2 is meant to provide, now for every measurable f , so Steps 1 and 3 complete the proof of Theorem 1.1 of [1] (Section 4).

The idea. In [1] the function is replaced by its averages f_h over the cells of $h\mathbb{Z}^d$. The averages destroy the oscillation of f inside a cell, and recovering the continuous form in the limit is what requires domination. We instead *sample* f at the points of a lattice. On each lattice the discrete theorem then compares sums of the two integrands, $(f(t) - f(s))^2 |s - t|^{-d-\alpha}$ and $(f(t) - f(s))^2 k(s, t)$, over pairs of lattice points s, t . Unlike with averages, the kernel $k(s, t)$ always meets the difference $f(t) - f(s)$ at the same pair of points. The lattice is then made random, by a shift $\eta \in [-\frac{1}{2}, \frac{1}{2}]^d$ and a linear map A close to the identity. Averaging the discrete inequality over the lattices reproduces both continuous forms, up to constants and up to pairs at distance $O(h)$. The random shift alone would sample only differences in directions of $h\mathbb{Z}^d$; the random distortion spreads the differences over all directions.

Related work. Chaker and Silvestre [2, Theorem 1.3] prove $\iint_{B_2 \times B_2} (u(x) - u(y))^2 K \geq c\lambda |u|_{\dot{H}^s(B_1)}^2$ for every u , under their Assumption 1.1: for every ball B and $x \in B$, the set of $z \in B$ with $K(x, z) \geq \lambda |x - z|^{-d-2s}$ has measure at least $\mu|B|$. They remark that their result implies that of [1]. A kernel satisfying (2) does satisfy Assumption 1.1, with $s = \alpha/2$, $\lambda = \Lambda^{-1}$ and $\mu = (\sin^2 \vartheta/16)^d$ (Lemma 9), so their theorem applies (Remark 10). Their method is different: an iteration on kernels with a covering argument, not a discretization. So, with Lemma A.1 of [1], the continuous conclusions of [1] were already available through [2]. The point of the present note is that the discretization route of [1], through their Theorem 1.3, can itself be completed. We are not aware of any earlier discussion of the gap.

Formal verification. Theorem A and every lemma below have been formalized and checked in Lean 4 with Mathlib (Section 5). The formalization also proves Theorem 1.3 of [1] itself.

2. THE GAP

We follow the proof of Lemma 3.7 of [1] (pp. 14–16). Let $f \in H_k(B^*)$. For $h \in (0, 1)$ let $f_h(x) = h^{-d} \int_{A_h(x) \cap B^*} f$ for $x \in h\mathbb{Z}^d \cap B^*$, where $A_h(x)$ is the cell of side h about x . By their Corollary 3.6 there are a constant C and a ϑ' -bounded configuration Γ_h , for some $\vartheta' > 0$, for which the discretized kernel ω_h^k satisfies the discrete bounds (8) of [1] on $h\mathbb{Z}^d$, with $R_0 = \sqrt{d}$ and $\Lambda = C$. Their Corollary 3.1 (Theorem 1.3 on $h\mathbb{Z}^d$) and Lemma 3.4 then give the discrete

inequality (15) of [1]:

$$\begin{aligned} c \sum_{\substack{x, y \in B \cap h\mathbb{Z}^d \\ |x-y| > \sqrt{d}h}} (f_h(x) - f_h(y))^2 \int_{A_h(x) \times A_h(y)} |s-t|^{-d-\alpha} d(s, t) \\ \leq \sum_{\substack{x, y \in B^* \cap h\mathbb{Z}^d \\ |x-y| > \sqrt{d}h}} (f_h(x) - f_h(y))^2 \int_{A_h(x) \times A_h(y)} k(s, t) d(s, t). \end{aligned}$$

The right-hand side is $\int g_h$, where $g_h(s, t) = (f_h(x) - f_h(y))^2 k(s, t)$ for $(s, t) \in A_h(x) \times A_h(y)$ with x, y as in the sum, and $g_h = 0$ otherwise. By Lebesgue differentiation (their Lemma A.2), $g_h \rightarrow g = (f(s) - f(t))^2 k(s, t) \mathbf{1}_{B^* \times B^*}$ almost everywhere. The proof then states, on p. 16:

“For the right hand side in (15) this implies with help of dominated convergence $[\int g_h \rightarrow \int g]$.”

No dominating function is given, and none is available from the hypothesis $f \in H_k(B^*)$.

For cells inside B^* , the natural upper bound on g_h comes from Jensen’s inequality,

$$(f_h(x) - f_h(y))^2 \leq h^{-2d} \int_{A_h(x) \times A_h(y)} (f(s') - f(t'))^2 d(s', t'),$$

together with the upper bound in (2), $k(s, t) \leq \Lambda |s-t|^{-d-\alpha} \lesssim |s'-t'|^{-d-\alpha}$. This bounds g_h by local averages of $(f(s') - f(t'))^2 |s'-t'|^{-d-\alpha}$. Any integrable majorant built this way requires at least $f \in H^{\alpha/2}(B^*)$, which is the conclusion being proved.

The hypothesis controls $(f(s) - f(t))^2 k(s, t)$, and this cannot be used here. The function g_h pairs the value $k(s, t)$ with differences of *averages* of f over the two cells, and the lower bound in (2) controls k only on the cones. The kernel may vanish at (s', t') while being of full size at (s, t) in the same pair of cells.

Nor can the step be rescued by approximating $f \in H_k(B^*)$ by smooth functions in the H_k norm: that would require the density statement of Theorem 1.4, whose proof uses Lemma 3.7. The final inequality of Lemma 3.7 is true (Corollary 8), but its proof in [1] is incomplete at this step.

3. PROOF OF THEOREM A

Throughout, $d \geq 1$, $|\cdot|$ is the Euclidean norm and $|\cdot|_\infty$ the maximum norm, and $|\cdot|$ also denotes Lebesgue measure. Balls $B_r(x)$ are open and $\bar{B}_r(x)$ closed. Integrals of $[0, \infty]$ -valued functions are Lebesgue integrals, and all identities and inequalities are between elements of $[0, \infty]$. We use the convention $0 \cdot \infty = 0$, and read the integrand of $|f|_{H^{\alpha/2}}^2$ as 0 on the diagonal.

The proof has three parts. Lemma 2 applies Theorem 1.3 of [1] on a single lattice. Proposition 6 averages over lattices: for every nonnegative measurable H , the average of $\sum H(Tz, Tz')$ is comparable to $h^{-2d} \iint H$, except that pairs at distance $O(h)$ are lost on the lower side, and no integrability of H is needed. Applied to the two integrands, this gives the inequality of Theorem A for each h , up to those short pairs. The only limit, $h \rightarrow 0$, is on the $H^{\alpha/2}$ side, where it is monotone. The H_k side, where the proof in [1] needed domination, involves no limit.

3.1. The discrete theorem.

Theorem 1.3 of [1]. Let Γ be a ϑ -bounded configuration and $\alpha \in (0, 2)$. Let $\omega: \mathbb{Z}^d \times \mathbb{Z}^d \rightarrow [0, \infty]$ satisfy $\omega(x, y) = \omega(y, x)$ and

$$\Lambda^{-1}(\mathbf{1}_{V\Gamma[x]}(y) + \mathbf{1}_{V\Gamma[y]}(x))|x - y|^{-d-\alpha} \leq \omega(x, y) \leq \Lambda|x - y|^{-d-\alpha}$$

for $|x - y| > R_0$, where $R_0 > 0$ and $\Lambda \geq 1$. There are $\kappa \geq 1$ and $c \geq 1$ such that for every $R > 0$, every $x_0 \in \mathbb{R}^d$ and every $f: B_{\kappa R}(x_0) \cap \mathbb{Z}^d \rightarrow \mathbb{R}$,

$$\sum_{\substack{x, y \in B_R(x_0) \cap \mathbb{Z}^d \\ |x - y| > R_0}} (f(x) - f(y))^2 |x - y|^{-d-\alpha} \leq c \sum_{\substack{x, y \in B_{\kappa R}(x_0) \cap \mathbb{Z}^d \\ |x - y| > R_0}} (f(x) - f(y))^2 \omega(x, y).$$

The constant c depends on Λ , ϑ , R_0 and d . It does not depend on ω and Γ .

The statement leaves the dependence of κ open. The proof in [1] gives $\kappa = (N - 1)\lambda$ (p. 29), and κ , like c , can be chosen depending only on Λ , ϑ , R_0 and d . We use Theorem 1.3 in this uniform form. So does the proof of Lemma 3.7 in [1], which fixes κ and then applies Corollary 3.1 to configurations and kernels that vary with h .

We take $R_0 = \frac{1}{2}$. Then the condition $|x - y| > R_0$ holds for all distinct lattice points, and the sums run over all pairs of distinct points, or all pairs, since diagonal terms vanish.

3.2. Random lattices. Fix

$$\varepsilon = \frac{1}{2} \sin(\vartheta/2), \quad \delta = \varepsilon/d.$$

For $Q = (Q_1, \dots, Q_d) \in (\mathbb{R}^d)^d$ let A_Q be the linear map $A_Q u = u + \sum_j u_j Q_j$, and let $\mathcal{Q}_\delta = \bar{B}_\delta(0)^d \subseteq (\mathbb{R}^d)^d$. For $Q \in \mathcal{Q}_\delta$, $|A_Q u - u| \leq \sum_j |u_j| \delta \leq d\delta|u| = \varepsilon|u|$. Hence

$$(*) \quad (1 - \varepsilon)|u| \leq |A_Q u| \leq (1 + \varepsilon)|u|, \quad (1 - \varepsilon)^d \leq |\det A_Q| \leq (1 + \varepsilon)^d,$$

the second because the singular values of A_Q lie in $[1 - \varepsilon, 1 + \varepsilon]$. Let $P = [-\frac{1}{2}, \frac{1}{2}]^d$. For $h > 0$, $Q \in \mathcal{Q}_\delta$ and $\eta \in P$, the lattice map is

$$T(z) = T_{h, Q, \eta}(z) = h A_Q(z + \eta) \quad (z \in \mathbb{R}^d),$$

an injective affine map of $\mathbb{R}^d$, which we evaluate at the points of $\mathbb{Z}^d$.

3.3. One lattice.

Lemma 1 (Cones). Let V be a double cone with axis v and apex angle $\theta \geq \vartheta$, and let V' be the double cone with the same axis and apex angle $\vartheta/2$. Let A be linear with $|Au - u| \leq \varepsilon|u|$ for all u , where $\varepsilon < \sin(\vartheta/2)$. Then $AV' \subseteq V$.

Proof. Let $w \in V'$, so w makes an angle less than $\vartheta/2$ with v or with $-v$. Since $|Aw - w| \leq \varepsilon|w|$ and $\varepsilon < 1$, $Aw \neq 0$, and the angle between Aw and w is at most $\arcsin \varepsilon < \vartheta/2$. By the triangle inequality for angles, Aw makes an angle less than $\vartheta \leq \theta$ with the same half-line, so $Aw \in V$. $\square$

Lemma 2 (The inequality for one lattice). Let κ_0 and c_0 be the constants of the uniform form of Theorem 1.3 of [1] for the parameters $\vartheta/2$, $\Lambda' = 2^{d+\alpha}\Lambda$ and $R_0 = \frac{1}{2}$. Let Γ be ϑ -bounded, let k be symmetric and satisfy (2), and let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be any function. Write

$$F(s, t) = (f(t) - f(s))^2 |s - t|^{-d-\alpha}, \quad G(s, t) = (f(t) - f(s))^2 k(s, t),$$

and $F_D = F \mathbf{1}_{D \times D}$, $G_D = G \mathbf{1}_{D \times D}$. Then for every ball $B = B_R(x_0)$, every $h > 0$, $Q \in \mathcal{Q}_\delta$ and $\eta \in P$, with $T = T_{h, Q, \eta}$ and $B^* = B_{3\kappa_0 R}(x_0)$,

$$\sum_{z, z' \in \mathbb{Z}^d} F_B(Tz, Tz') \leq 2^{d+\alpha} c_0 \sum_{z, z' \in \mathbb{Z}^d} G_{B^*}(Tz, Tz').$$

Proof. Write $A = A_Q$.

The pulled-back configuration. Let $\Gamma'(z)$ be the double cone with the axis of $\Gamma(Tz)$ and apex angle $\vartheta/2$, for every $z \in \mathbb{R}^d$. All its apex angles equal $\vartheta/2$, so Γ' is $\vartheta/2$ -bounded. If $z' \in V^{\Gamma'}[z]$, then $Tz' - Tz = hA(z' - z) \in \Gamma(Tz)$ by Lemma 1, since double cones are invariant under positive dilations. Hence $\mathbf{1}_{V^{\Gamma'}[z]}(z') \leq \mathbf{1}_{V^{\Gamma}[Tz]}(Tz')$.

The pulled-back kernel. Let $\omega(z, z') = h^{d+\alpha}k(Tz, Tz')$, which is symmetric. By (*), $h(1 - \varepsilon)|z - z'| \leq |Tz - Tz'| \leq h(1 + \varepsilon)|z - z'|$. With $\varepsilon \leq \frac{1}{2}$, (2) gives, for $z \neq z'$,

$$\omega(z, z') \leq \Lambda(1 - \varepsilon)^{-d-\alpha}|z - z'|^{-d-\alpha} \leq \Lambda'|z - z'|^{-d-\alpha},$$

and

$$\omega(z, z') \geq \Lambda^{-1}(1 + \varepsilon)^{-d-\alpha}(\mathbf{1}_{V^{\Gamma'}[z]}(z') + \mathbf{1}_{V^{\Gamma'}[z']}(z))|z - z'|^{-d-\alpha},$$

with $\Lambda^{-1}(1 + \varepsilon)^{-d-\alpha} \geq \Lambda'^{-1}$. These bounds hold at every pair of lattice points, because (2) holds at every pair of points; no measurability of Γ enters. So (Γ', ω) satisfies the hypotheses of Theorem 1.3 of [1] with $\vartheta/2$, Λ' and $R_0 = \frac{1}{2}$.

The balls. Let $\xi = T^{-1}x_0$ and $r = R/(h(1 - \varepsilon))$. If $Tz \in B$ then $|z - \xi| \leq |Tz - x_0|/(h(1 - \varepsilon)) < r$, so $z \in B_r(\xi)$. If $z \in B_{\kappa_0 r}(\xi)$ then $|Tz - x_0| < h(1 + \varepsilon)\kappa_0 r = \kappa_0 R(1 + \varepsilon)/(1 - \varepsilon) \leq 3\kappa_0 R$, so $Tz \in B^*$.

Conclusion. Apply Theorem 1.3 of [1] to $f \circ T$ on $B_r(\xi)$. Using $|Tz - Tz'|^{-d-\alpha} \leq 2^{d+\alpha}h^{-d-\alpha}|z - z'|^{-d-\alpha}$ on the left,

$$\begin{aligned} \sum_{z, z'} F_B(Tz, Tz') &\leq 2^{d+\alpha}h^{-d-\alpha} \sum_{z, z' \in B_r(\xi)} (f(Tz) - f(Tz'))^2 |z - z'|^{-d-\alpha} \\ &\leq 2^{d+\alpha}c_0 h^{-d-\alpha} \sum_{z, z' \in B_{\kappa_0 r}(\xi)} (f(Tz) - f(Tz'))^2 \omega(z, z'). \end{aligned}$$

The last sum is $h^{d+\alpha} \sum_{z, z' \in B_{\kappa_0 r}(\xi)} G(Tz, Tz')$, and every such pair has $Tz, Tz' \in B^*$. $\square$

3.4. Averaging over lattices. For measurable $H: \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, \infty]$ put

$$L_h(H) = \int_{\mathcal{Q}_\delta} \int_P \sum_{z, z' \in \mathbb{Z}^d} H(T_{h, Q, \eta} z, T_{h, Q, \eta} z') d\eta dQ.$$

Here dQ is Lebesgue measure on $(\mathbb{R}^d)^d$, and the integrand is jointly measurable in (Q, η) .

Without the distortion, that is for $Q = 0$, the average over η alone is $h^{-d} \sum_{w \neq 0} \int H(s, s + hw) ds$ (Lemma 3), which sees H only at offsets in $h\mathbb{Z}^d$. Averaging over Q replaces each offset hw by a ball about it of radius comparable to $\delta h|w|_\infty$ (Lemma 4), and these balls cover all offsets of length at least $C_0 h$ with bounded overlap (Lemma 5).

Lemma 3 (Tiling). *Let H be measurable, nonnegative and zero on the diagonal. For $h > 0$ and $Q \in \mathcal{Q}_\delta$,*

$$\int_P \sum_{z, z'} H(Tz, Tz') d\eta = |\det(hA_Q)|^{-1} \sum_{w \in \mathbb{Z}^d \setminus \{0\}} \int_{\mathbb{R}^d} H(s, s + hA_Q w) ds.$$

Proof. Write $z' = z + w$. The term $w = 0$ vanishes. For fixed w , the translates $z + P$, $z \in \mathbb{Z}^d$, tile $\mathbb{R}^d$, so $\sum_z \int_P H(hA(z + \eta), hA(z + \eta) + hAw) d\eta = \int_{\mathbb{R}^d} H(hAu, hAu + hAw) du$. The substitution $s = hAu$ finishes the proof. All terms are nonnegative, so sums and integrals may be interchanged. $\square$

Lemma 4 (Averaging a column). *Let $g: \mathbb{R}^d \rightarrow [0, \infty]$ be measurable and $w \in \mathbb{Z}^d \setminus \{0\}$, with $m = |w|_\infty$. Then*

$$|B_{\delta/d}|^{d-1} m^{-d} \int_{\bar{B}_{\delta m/d}(w)} g \leq \int_{Q_\delta} g(A_Q w) dQ \leq |B_\delta|^{d-1} m^{-d} \int_{\bar{B}_{\delta m}(w)} g.$$

Proof. Choose j with $|w_j| = m$. Fix the columns Q_i , $i \neq j$, and put $b = w + \sum_{i \neq j} w_i Q_i$, so that $A_Q w = b + w_j Q_j$. The substitution $y = b + w_j Q_j$ gives $\int_{\bar{B}_\delta(0)} g(b + w_j Q_j) dQ_j = m^{-d} \int_{\bar{B}_{\delta m}(b)} g$.

For the upper bound, $|b - w| \leq (d-1)\delta m$, so $\bar{B}_{\delta m}(b) \subseteq \bar{B}_{\delta m}(w)$. Integrating over the other $d-1$ columns gives the factor $|B_\delta|^{d-1}$.

For the lower bound, restrict the other columns to $\bar{B}_{\delta/d}(0)$. Then $|b - w| \leq (d-1)\delta m/d$ and $\bar{B}_{\delta m}(b) \supseteq \bar{B}_{\delta m/d}(w)$, and the restricted columns contribute $|B_{\delta/d}|^{d-1}$. $\square$

Lemma 5 (Counting). *For $y \in \mathbb{R}^d$ put $\rho^+(y) = \sum_{w \neq 0} |w|_\infty^{-d} \mathbf{1}[|y - w| \leq d\delta|w|_\infty]$ and $\rho^-(y) = \sum_{w \neq 0} |w|_\infty^{-d} \mathbf{1}[|y - w| \leq \delta|w|_\infty/d]$, the sums over $w \in \mathbb{Z}^d \setminus \{0\}$. Then*

- (1) $\rho^+(y) \leq 6^d$ for all y ;
- (2) $\rho^-(y) \geq (\delta/(3d^2))^d$ whenever $|y| \geq C_0 := 2d^3/\delta$.

Proof. (1) If w contributes, then $|y - w|_\infty \leq |y - w| \leq d\delta|w|_\infty \leq \frac{1}{2}|w|_\infty$. Hence $\frac{1}{2}|w|_\infty \leq |y|_\infty \leq \frac{3}{2}|w|_\infty$. Since $|w|_\infty \geq 1$, this forces $|y|_\infty \geq \frac{1}{2}$. The contributing w lie in the cube $\{|w - y|_\infty \leq |y|_\infty\}$, which contains at most $(2|y|_\infty + 1)^d \leq (4|y|_\infty)^d$ lattice points. Each has weight $|w|_\infty^{-d} \leq (\frac{3}{2}|y|_\infty)^d$. So $\rho^+(y) \leq 6^d$.

(2) Let $M = |y|_\infty \geq |y|/\sqrt{d} \geq |y|/d \geq 2d^2/\delta$, and $r = \delta M/(2d^2) \geq 1$. Since $\delta \leq 1$, also $r \leq M/2$. Consider $w \in \mathbb{Z}^d$ with $|w - y|_\infty \leq r$. Then $\frac{1}{2}M \leq |w|_\infty \leq \frac{3}{2}M$, in particular $w \neq 0$, and

$$|y - w| \leq \sqrt{d}r \leq dr = \delta M/(2d) \leq \delta|w|_\infty/d,$$

so w contributes at least $(\frac{3}{2}M)^{-d}$. Each coordinate interval $[y_i - r, y_i + r]$ contains at least $\lfloor 2r \rfloor \geq r$ integers, since $r \geq 1$. So there are at least r^d such w , and $\rho^-(y) \geq r^d (\frac{3}{2}M)^{-d} = (\delta/(3d^2))^d$. $\square$

Proposition 6 (The lattice average). *Let $H: \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, \infty]$ be measurable and zero on the diagonal. For every $h > 0$,*

$$K_- h^{-2d} \iint_{|t-s| \geq C_0 h} H(s, t) ds dt \leq L_h(H) \leq K_+ h^{-2d} \iint H(s, t) ds dt,$$

with $K_+ = 6^d |B_\delta|^{d-1} (1 - \varepsilon)^{-d}$, $K_- = (\delta/(3d^2))^d |B_{\delta/d}|^{d-1} (1 + \varepsilon)^{-d}$ and C_0 as in Lemma 5.

Proof. By Lemma 3, (*) and Tonelli's theorem,

$$L_h(H) \leq h^{-d} (1 - \varepsilon)^{-d} \sum_{w \neq 0} \int_{\mathbb{R}^d} \int_{Q_\delta} H(s, s + hA_Q w) dQ ds.$$

Apply Lemma 4 with $g(y) = H(s, s + hy)$, and then sum over w :

$$L_h(H) \leq h^{-d} (1 - \varepsilon)^{-d} |B_\delta|^{d-1} \iint H(s, s + hy) \rho^+(y) dy ds.$$

By Lemma 5(1) and the substitution $t = s + hy$, for which $\int H(s, s + hy) dy = h^{-d} \int H(s, t) dt$, this is at most $K_+ h^{-2d} \iint H$.

The lower bound is the same computation, with the lower halves of $(*)$ and Lemma 4. Discarding $|y| < C_0$ and using Lemma 5(2) gives

$$L_h(H) \geq h^{-d}(1+\varepsilon)^{-d}|B_{\delta/d}|^{d-1}\left(\frac{\delta}{3d^2}\right)^d \int_{\mathbb{R}^d} \int_{|y| \geq C_0} H(s, s+hy) dy ds,$$

and the substitution $t = s + hy$ turns $\{|y| \geq C_0\}$ into $\{|t - s| \geq C_0 h\}$. $\square$

3.5. Conclusion.

Proof of Theorem A. With κ_0, c_0 as in Lemma 2, put $\kappa = 3\kappa_0$ and $B^* = B_{\kappa R}(x_0)$. Since f and k are measurable, F_B and G_{B^*} are measurable, and both vanish on the diagonal, whatever the value of k there. Lemma 2 holds for every (h, Q, η) . Integrating it over $\eta \in P$ and $Q \in \mathcal{Q}_\delta$ gives $L_h(F_B) \leq 2^{d+\alpha} c_0 L_h(G_{B^*})$. By Proposition 6,

$$K_- \iint_{|t-s| \geq C_0 h} F_B \leq 2^{d+\alpha} c_0 K_+ \iint G_{B^*} \quad (h > 0).$$

As $h \downarrow 0$ the sets $\{|t - s| \geq C_0 h\}$ increase to the complement of the diagonal, and the diagonal is a null set. By monotone convergence,

$$|f|_{H^{\alpha/2}(B)}^2 = \iint F_B \leq \frac{2^{d+\alpha} c_0 K_+}{K_-} \iint G_{B^*} = C |f|_{H_k(B^*)}^2.$$

The constants $\varepsilon, \delta, K_\pm, C_0$ depend only on d and ϑ , and κ_0, c_0 only on d, ϑ, Λ and α . $\square$

Remark 7. The constants κ and C of Theorem A can be chosen independently of $\alpha \in (0, 2)$. Indeed, so can the constants of the uniform form of Theorem 1.3 (Section 3.1). Its proof produces a factor $\lambda^{d+\alpha}$, where $\lambda \geq R_0$ is the constant of Theorem 5.15 of [1]. Since λ may be taken ≥ 1 , this is at most λ^{d+2} . Then use $\Lambda' = 2^{d+2}\Lambda$ in Lemma 2, and $2^{d+\alpha} \leq 2^{d+2}$. Both uniform statements are formalized (Section 5).

4. CONSEQUENCES

A configuration is ϑ -admissible in [1] if it is ϑ -bounded and $\{(x, y) : y - x \in \Gamma(x)\}$ is Borel (condition (M)). Theorem A does not use (M).

Corollary 8. *Lemma 3.7, Theorem 1.1, and the comparability statements of Theorem 1.4 of [1] hold as stated.*

Proof. In the proof of Lemma 3.7, the passage in Section 2 is meant to establish $c|f|_{H^{\alpha/2}(B)}^2 \leq |f|_{H_k(B^*)}^2$ for all $f \in H_k(B^*)$ and every ball, with c independent of the ball. Theorem A gives this for every $f \in L^2(B^*)$: take a measurable representative, extended by zero, and note that both sides depend only on the values of f on B^* . The rest of the proof of Lemma 3.7 in [1] is the application of their Lemma A.1, which removes the enlargement and is unaffected by the gap. Theorem 1.1 follows from Lemma 3.7 as in [1]. So do the comparabilities in Theorem 1.4: on $\mathbb{R}^d$ by letting the radius tend to infinity, and on bounded Lipschitz domains through Lemma A.1. $\square$

The dependence on α_0 asserted in Lemma 3.7 and Theorem 1.1 follows from Remark 7 and the corresponding statement of Lemma A.1.

4.1. The route through Chaker and Silvestre. Chaker and Silvestre [2, Assumption 1.1] ask for $\mu \in (0, 1)$ and $\lambda > 0$ such that for every ball B and every $x \in B$, $|\{z \in B : K(x, z) \geq \lambda|x - z|^{-d-2s}\}| \geq \mu|B|$.

Lemma 9. *Let Γ be ϑ -bounded, and let k be symmetric and satisfy (2). Then for every ball $B = B_r(p)$ and every $x \in B$,*

$$|\{z \in B : k(x, z) \geq \Lambda^{-1}|x - z|^{-d-\alpha}\}| \geq \left(\frac{\sin^2 \vartheta}{16}\right)^d |B|.$$

That is, k satisfies Assumption 1.1 of [2] with $s = \alpha/2$, $\lambda = \Lambda^{-1}$ and $\mu = (\sin^2 \vartheta/16)^d$.

Proof. Put $\sigma = \sin \vartheta$. Let v be the axis of $\Gamma(x)$, with its sign chosen so that $\langle v, p - x \rangle \geq 0$. Since the apex angle of $\Gamma(x)$ is at least ϑ , $\Gamma(x)$ contains the half-cone $W = \{u \neq 0 : \angle(u, v) < \vartheta\}$. For $u \in \mathbb{R}^d$ put $\phi(u) = \langle v, u \rangle \sigma - |u - \langle v, u \rangle v| \cos \vartheta$. Then $\phi(u) = |u| \sin(\vartheta - \angle(u, v))$ for $u \neq 0$, so $\phi > 0$ exactly on W . By the Cauchy–Schwarz inequality for the pair $(\sigma, \cos \vartheta)$, ϕ is 1-Lipschitz, and $\phi(u + tv) = \phi(u) + t\sigma$.

Let $a = r\sigma/4$, $\tau = \sigma^2/8$, $\rho = r\sigma^2/16$ and $q = x + av + \tau(p - x)$. We show that $B_\rho(q) \subseteq B$ and $B_\rho(q) - x \subseteq W$.

First, $q - p = av - (1 - \tau)(p - x)$, and $\langle v, p - x \rangle \geq 0$, $|p - x| < r$, so

$$|q - p|^2 \leq a^2 + (1 - \tau)^2 r^2 = (r - \rho)^2 - \frac{r^2 \sigma^2 (16 - 3\sigma^2)}{256} \leq (r - \rho)^2.$$

Hence $|q - p| + \rho \leq r$, and $B_\rho(q) \subseteq B$.

Second, $\phi(q - x) = \phi(\tau(p - x)) + a\sigma \geq -\tau|p - x| + a\sigma > -\tau r + a\sigma = r\sigma^2/8$, using $\phi(0) = 0$. For $z \in B_\rho(q)$, $\phi(z - x) \geq \phi(q - x) - |z - q| > r\sigma^2/8 - \rho > 0$, so $z - x \in W \subseteq \Gamma(x)$.

For $z \in B_\rho(q)$, therefore, $z \in B$ and $z \in V^\Gamma[x]$, and (2) gives $k(x, z) \geq \Lambda^{-1}|x - z|^{-d-\alpha}$. Finally $|B_\rho(q)| = (\rho/r)^d |B| = (\sigma^2/16)^d |B|$. $\square$

Remark 10. By Lemma 9, [2, Theorem 1.3] applies to every kernel satisfying (2). Rescaling $B_R(x_0)$ to $B_1(0)$ preserves (2), since double cones are invariant under positive dilations, so [2, Theorem 1.3] gives the enlarged-ball inequality of Theorem A with $\kappa = 2$.

5. FORMALIZATION

The argument above is checked in Lean 4 with Mathlib, as part of a formalization of [1]. The repository is <https://github.com/dbenbenn/quadratic-forms-sobolev-lean>. The headline results are stated in Lean exactly as above, with ϑ -boundedness as in Definition 2.1 of [1] (QFS.IsThetaBounded):

Theorem A	QFS.GapNote.theorem_A
Remark 7	QFS.GapNote.remark_7
Lemma 9	QFS.GapNote.lemma_9
Theorem 1.3 of [1]	QFS.Paper.theorem_1_3

The first three are in the file `GapNote.lean`, the last in `Paper.lean`, and the lemmas in the directory `RandomLattice`, all under `QuadraticFormsSobolev`:

Lemma 1	QFS.mem_doubleCone_of_near
Lemma 2	QFS.latticeSum_ball_le
Lemma 3	QFS.lintegral_latticeSum_le, QFS.le_lintegral_latticeSum
Lemma 4	QFS.lintegral_colBox_le, QFS.le_lintegral_colBox
Lemma 5	QFS.tsum_count_le, QFS.le_tsum_count
Proposition 6	QFS.latAvg_le, QFS.le_latAvg
Uniform Theorem 1.3	QFS.theoremOneThree_uniform

These are the forms the Lean proof uses, not transcriptions. The Lean form of Lemma 2 assumes only that ϑ is a lower bound for the apex angles (`QFS.ApexLowerBound`), and allows $\alpha \geq 0$. The two tiling inequalities combine Lemma 3 with the bound $(*)$ on the determinant. Some constants differ harmlessly: 4^d in Lemma 5(1) and $(\delta/(4d^2))^d$ in Lemma 5(2).

The formalization also proves Theorem 1.3 of [1] in full, in the α -uniform form of Remark 7, and Theorem 1.1 with the α_0 clause. The statement of Theorem 1.3 is correct as printed. A few steps of its printed proof needed local repairs in the formalization; they are independent of the gap discussed here and are not treated in this note. For the step removing the enlargement, the formalization gives a proof of Lemma A.1 that avoids the Whitney decomposition, with Dyda’s inequality proved rather than quoted. All results depend only on the standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

REFERENCES

- [1] K.-U. Bux, M. Kassmann, and T. Schulze, *Quadratic forms and Sobolev spaces of fractional order*, Proc. London Math. Soc. (3) **119** (2019), 841–866, [doi:10.1112/plms.12246](https://doi.org/10.1112/plms.12246). Cited from [arXiv:1707.09277v1](https://arxiv.org/abs/1707.09277v1), whose numbering and page numbers we use; we were unable to consult the journal version.
- [2] J. Chaker and L. Silvestre, *Coercivity estimates for integro-differential operators*, Calc. Var. Partial Differential Equations **59** (2020), Paper No. 106, [doi:10.1007/s00526-020-01764-y](https://doi.org/10.1007/s00526-020-01764-y). Cited from [arXiv:1904.13014v1](https://arxiv.org/abs/1904.13014v1), whose numbering we use; we were unable to consult the journal version.
- [3] B. Dyda, *On comparability of integral forms*, J. Math. Anal. Appl. **318** (2006), 564–577, [doi:10.1016/j.jmaa.2005.06.021](https://doi.org/10.1016/j.jmaa.2005.06.021).

About this note. This note was written by Claude, an AI model made by Anthropic. The mathematics in it was developed and machine-checked in the Lean formalization cited in Section 5.

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